

Practical 7 - Competing risks 2

Survival Analysis

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Datasets required: `aaatrial.RData`

R packages required: `survival`, `cmprsk`

```
library(survival)
library(cmprsk)
```

Introduction

In this practical we will use the `aaatrial` data, as in Practical 6. This is a dataset from a randomised trial of screening for abdominal aortic aneurysm (AAA), in which older men were randomised to be invited to be screened for AAA (invited group) or not (control group). The main outcome of interest was deaths relating to AAA (including deaths following AAA repair operations), but men in the study also died from other causes. There are 13,562 people in the data.

If an AAA was found at screening, a repair operation was carried out, essentially removing the risk of future death from AAA. However, the operation itself carries around a 5% risk of mortality. Undetected AAAs may rupture at any time, resulting in either an emergency repair operation (which carries around a 40% risk of mortality) or death. Participants were recruited to the trial between 1997 and 1999 and followed up until the end of 2007.

The key variables are described below.

Variable	Description
<code>id</code>	Unique identifier for each participant
<code>group</code>	Randomisation group: 0=Control, 1=Invited
<code>dateran</code>	Date of randomisation
<code>aaadeath</code>	AAA-death indicator: 0=Censored or non-AAA-death, 1=AAA-death
<code>allddeath</code>	All-cause mortality: 0=Censored, 1=Death (any cause)

Variable	Description
deathtype	0=Censored; 1=non-AAA death; 2=AAA-death
timeout	Date of study exit
dob	Date of birth

This is already an R dataset, and the dates have been formatted for you as dates.

Aims

The aims of this session are:

- Fit and interpret results from cause-specific hazard models.
- Fit and interpret results from subdistribution hazard models.
- Estimate cumulative incidence functions based on cause-specific hazard models and based on a subdistribution hazard model.

Question 1

- Load the data. Create a variable called `time`, which is the follow-up time variable you created in Practical 6, Question 1.
- Fit cause-specific Cox models for AAA-death and non-AAA-death with `group` as the explanatory variable. This can be done using the code below. Note that the `id` option is required when using `coxph` for competing risks.

```
cox.cs<-coxph(Surv(time,as.factor(deathtype))~group,data=aaatrial,id=id)
```

- Write down the form of the hazard models and interpret the cause-specific hazard ratios.
- An equivalent way of fitting these models is by using `coxph` for AAA-deaths and non-AAA-deaths separately, like this:

```
cox.AAA<-coxph(Surv(time,deathtype==2)~group,data=aaatrial)
# cox.AAA<-coxph(Surv(time,aaadeath)~group,data=aaatrial)#same
cox.nonAAA<-coxph(Surv(time,deathtype==1)~group,data=aaatrial)
```

- How is the competing event treated in the fitting of the cause-specific Cox models?

Question 2

- In this question we will estimate the cumulative incidences of AAA-death and non-AAA-death based on the cause-specific hazard models fitted in question 1. After fitting the cause-specific Cox models as in `cox.cs` (using `deathtype` as a factor variable) we can estimate cumulative incidences as follows:

```
cinc.cs <- survfit(cox.cs,newdata = data.frame(group=c(0,1)))
summary(cinc.cs, times=c(1,5,10))

plot(cinc.cs,lwd=2,lty=c(1,2,1,2),col=c("black","black","grey","grey"))
legend("topleft",c("Control, Non-AAA-death","Invited, Non-AAA-death",
                  "Control, AAA-death","Invited, AAA-death"),
      lwd=2,lty=c(1,2,1,2),col=c("black","black","grey","grey"))
```

- What are the estimated cumulative incidences for AAA-death and for non-AAA-death at time 5 for the two treatments.

Question 3

- Create a new variable that is the participant's age in years at the time of randomisation, using the date of birth (`dob`) and date of randomisation (`dateran`).
- Fit cause-specific Cox models for AAA-death and non-AAA-death with group and age as the explanatory variables.
- Write down the forms of the hazard models and interpret the cause-specific hazard ratios.

Question 4

Based on the cause-specific hazard models fitted in question 3, obtain the estimated cumulative incidences for AAA-death and for non-AAA-death at time 5 for the following four individuals:

Person	Group	Age
1	0	65
2	1	65
3	0	70
4	1	70

Question 5

- In this question we will fit Fine and Gray models (i.e. subdistribution hazard models) for AAA-death, using group as the explanatory variable. This can be done using the `crr` function in the `cmprsk` package as follows:

```
crr.AAA <- crr(aaatrial$time, aaatrial$deathtype,  
              cov1=aaatrial$group, failcode=2)
```

- Interpret the subdistribution hazard ratios, and think about how the interpretation differs from that of the cause-specific hazard ratios.

Question 6

- Based on the Fine and Gray model from Question 5, obtain estimates of the cumulative incidences of AAA-death in the two groups using the code below.
- What is the estimated cumulative incidence up to 5 years.

```
cinc.fg.AAA <- predict.crr(crr.AAA, cov1 = c(0,1))  
#Note:  
#Column 1 is time  
#Column 2 is cumulative incidence for AAA in group 0  
#Column 3 is cumulative incidence for AAA in group 1  
head(cinc.fg.AAA)
```

Question 7

Using a plot, compare the cumulative incidences of non-AAA-death in the two treatment groups based on three estimation methods: (i) the non-parametric Aalen-Johansen approach, (ii) cause-specific hazard models, (iii) a subdistribution (Fine and Gray) model.

Have a go at making the plot yourself, or see the example code in the solutions.

Explain why you get different estimates using the three approaches.